Bhavya Gupta

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Research Highlights

I am a physics graduate student at the University of Minnesota, Twin Cities, interested in developing software for rapid transient detection and low-latency alert generation in time-domain astronomy. My research background includes machine-learning methods for gravitational-wave detection, with an emphasis on accelerating searches and enabling multi-messenger astronomy. I am particularly interested in building scalable, real-time computational systems that identify astrophysical transients and facilitate timely follow-up observations across multiple messengers to answer fundamental questions about our universe.

Education

- **University of Minnesota, Twin Cities** — *August, 2026 - Present*
  Ph.D., Physics — Minneapolis, MN
  - Advisor: Professor Michael W. Coughlin
- **University of California, San Diego** — *September 2020 - June 2024*
  B.S., Physics with Specialization in Astrophysics; Minor: Mathematics — La Jolla, CA
  - Department Honors: High Distinction
  - Honors Thesis: *Probing Dark Matter Subhalo Impacts on Stellar Streams with Graph Neural Networks and Normalizing Flows*

Research Experience

- **University of Minnesota, Twin Cities** — *August 2026 - Present*
  Graduate Research Assistant — Minneapolis, MN
  - Advisor: Professor Michael W. Coughlin
  - Developing scalable computational and machine-learning methods for rapid astrophysical transient detection and low-latency alert generation in the gravitational-wave domain, enabling timely multi-messenger follow-up observations.
- **Massachusetts Institute of Technology** — *July 2025 - July 2026*
  Accelerated AI Algorithms for Data-Driven Discovery (A3D3) Post-baccalaureate Fellow — Cambridge, MA
  - Advisors: Professor Erik Katsavounidis, Dr. Deep Chatterjee
  - Developing machine-learning frameworks for low-latency detection and parameter estimation of binary neutron star mergers in LIGO-Virgo-KAGRA data by extending `Aframe` and `AMPLFI` to improve sensitivity to low-SNR inspiral signals and enable rapid multi-messenger follow-up.
- **University of California San Diego** — *January 2023 - June 2025*
  Undergraduate Student Researcher — La Jolla, CA
  - Advisors: Professor Javier Duarte, Professor Tongyan Lin
  - Modeled dark matter (DM) subhalo interactions with stellar streams by adapting Erkal (2015)'s analytic framework with `Gala` and `Galpy`, developing an MCMC pipeline to infer subhalo impact parameters, and extending the analysis to complex, non-analytic scenarios using Graph Neural Networks (GNNs) and Normalizing Flows for Simulation-Based Inference (SBI); benchmarked inference accuracy and computational efficiency on distributed CPU/GPU clusters at the San Diego Supercomputer Center.
- **University of California San Diego** — *September 2022 - December 2022*
  Undergraduate Student Researcher — La Jolla, CA
  - Advisor: Professor Alison Coil
  - Investigated supermassive black hole (SMBH) outflows and their role in galaxy evolution, performing spectral analysis and two-component Gaussian modeling of emission-line kinematics using Python and IDL's IFSFIT, and identifying the need for a three-component model to better capture complex outflow structures.
- **Carnegie Observatories** — *September 2022 - December 2022*
  Carnegie Astrophysics Summer Student Internship (CASSI) — Pasadena, CA
  - Advisor: Dr. Peter Senchyna
  - Investigated metal-poor, star-forming dwarf galaxies ($z < 1$) as analogs of high-redshift systems ($z > 6$), analyzing SOFIA spectroscopic data cubes with `SOSPEX` to extract [O III] and [C II] emission-line diagnostics, compare line ratios and morphology against star formation and metallicity indicators and the Dwarf Galaxy Survey, and found strong low-ionization [C II] emission from photodissociation regions (PDRs), highlighting differences in interstellar medium structure between local analogs and early galaxies.
- **Center for Matter at Atomic Pressures (CMAP), University of Rochester** — *August 2021*
  CMAP Undergraduate Summer School - Summer Student — Rochester, NY (Remote)
  - Advisor: Professor Pierre Gourdain
  - Analyzed plasma simulations using scientific Python packages, investigating hydrodynamics, magnetohydrodynamics, and particle dynamics to study how planetary density and particle size influence planet formation and atmospheric evaporation.

Publications

A=Abstract, C=Conference, J=Journal, T=Thesis

[J.1] Gupta, B., Chatterjee, D., Benoit, W., Marx, E., Reissel, C., Tsukamoto, S., Yoon, K., Coughlin, M. W., Harris, P., & Katsavounidis, E. (2026). **AI-enabled Gravitational-Wave Searches for Binary Neutron Stars at Optimal Sensitivity**. arXiv preprint arXiv:2607.01372 [astro-ph.HE]. Submitted to *Physical Review Letters*.

[A.1] Gupta, B., ML4GW Team (2026). **Advancing Low-Latency Gravitational-Wave Detection through Machine Learning**. *Bulletin of the American Astronomical Society*, 58(4).

[A.2] Gupta, B., Lin, T., & Sholapurkar, M. (2024). **Probing Dark Matter Subhalo Impacts on Stellar Streams with Graph Neural Networks**. *Bulletin of the American Astronomical Society*, 56(2).

[A.3] Gupta, B., & Senchyna, P. (2023). **Analyzing Far-IR [O III] and [C II] Emission Using SOFIA in Highly-Ionized Local Galaxies**. *Bulletin of the American Astronomical Society*, 55(2).

[T.1] Gupta, B. (2024). **Probing Dark Matter Subhalo Impacts on Stellar Streams with Graph Neural Networks and Normalizing Flows**. Undergraduate Honors Thesis, Department of Physics, University of California, San Diego.

Presentations & Posters

O=Oral, P=Poster

[O.1] 248th Meeting of the American Astronomical Society, Pasadena, CA *June 2026*

- *Advancing Low-Latency Gravitational-Wave Detection through Machine Learning*

[O.2] NSF HDR Ecosystem Conference, Columbus, OH *September 2025*

- *Accelerating Gravitational-Wave Astronomy with Machine Learning*

[P.1] NSF HDR Ecosystem Conference, Columbus, OH *September 2025*

- *Accelerating Gravitational-Wave Astronomy with Machine Learning*

[P.2] 243rd Meeting of the American Astronomical Society, New Orleans, LA *January 2024*

- *Probing Dark Matter Subhalo Impacts on Stellar Streams with Graph Neural Networks*

[P.3] Undergraduate Summer Research Award Symposium, UC, San Diego, La Jolla, CA *September 2023*

- *Simulating the Effects of Dark Matter Subhalos on Stellar Streams*

[P.4] 241st Meeting of the American Astronomical Society, Seattle, WA *January 2023*

- *Analyzing [O III] and [C II] Emission in Highly Ionized Local Systems*

[O.3] CASSI Summer Student Research Symposium, Pasadena, CA *August 2022*

- *Analyzing Far-IR [O III] and [C II] Emission Using SOFIA in Highly-Ionized Local Galaxies*

[P.5] CASSI Summer Student Research Symposium, Pasadena, CA *August 2022*

- Poster presentation: *Analyzing [O III] and [C II] Emission in Highly Ionized Local Systems*
- Received Best Poster Presentation Award.

Honors & Awards

- **A3D3 Post-baccalaureate Fellowship** *2025*
  Accelerated AI Algorithms for Data-Driven Discovery (A3D3)
  ◦ Selected for a fully funded, one-year research fellowship at the intersection of artificial intelligence, machine learning, and astrophysics.
- **Physical Sciences Dean's Undergraduate Award for Excellence** *2024*
  University of California, San Diego
  ◦ Recognized for academic excellence and promise as a researcher in the physical sciences; received a $1,000 award.
- **Revelle Provost Honors** *2020–2024*
  Revelle College, University of California, San Diego
- **School of Physical Sciences Undergraduate Summer Research Award (USRA)** *2023*
  University of California, San Diego
  ◦ Awarded $7,500 to conduct faculty-mentored summer research on machine-learning methods for studying dark matter subhalo impacts on stellar streams.
- **Society of Physics Students Outstanding Chapter Award** *2023*
  Society of Physics Students, National Office
  ◦ Recognized for contributions to strengthening the physics community through student engagement and chapter activities.
- **Pratibha Scholarship** *2021–2024*
  Aditya Birla Group
  ◦ Received $92,000 in scholarship support over three years toward undergraduate education through the Aditya Birla Group Employees' Children's Scholarship Plan.

- **CASSI Summer Student Poster Presentation Award** *2022*
  Carnegie Astrophysics Summer Student Internship
  ◦ Received the Best Poster Presentation Award for research on far-infrared [O III] and [C II] emission in nearby galaxies.
- **Carnegie Astrophysics Summer Student Internship (CASSI)** *2022*
  Carnegie Observatories
  ◦ Received a $6,000 research stipend to investigate far-infrared emission from nearby star-forming dwarf galaxies and its implications for metal-poor galaxies in the early Universe.
- **NSF CMAP Summer School Award** *2021*
  Center for Matter at Atomic Pressures (CMAP)
  ◦ Received a $300 stipend through a competitive summer school program in computational and applied physics.

Grants

- **Revelle Provost Academic Conference Support** *2023*
  Revelle College, University of California, San Diego
  ◦ Received a $300 travel grant to support attendance at the 243rd American Astronomical Society Meeting in New Orleans, LA.

Skills & Hobbies

- **Programming Languages:** Python, MATLAB, Mathematica
- **Scientific Computing & Data Analysis:** NumPy, SciPy, Pandas, Matplotlib, Numba, Astropy, Scikit-learn
- **Machine Learning & AI:** PyTorch, PyTorch Lightning, Keras, CUDA
- **Scientific Software & Typesetting:** LaTeX, Mathematica
- **Computing Environments & Tools:** Linux, Unix, HTCondor, Slurm, Kubernetes, Git
- **Productivity & Creative Software:** Microsoft Excel, Word, Office 365, Teams, Google Workspace, Adobe Photoshop, Adobe Lightroom
- **Languages:** English (Native), Hindi (Native), Spanish (Intermediate), Thai (Beginner)
- **Hobbies:** Photography, Painting, Badminton, Cricket, Biking

Leadership Experience

- **President** *Mar 2023 – Jun 2024*
  Astronomy Club at UC San Diego, La Jolla, CA
  ◦ Revived and re-registered the club, growing its community to 75 members, leading an 8-member board, and securing $1,500 in Student Success Center funding.
  ◦ Organized three on-campus and two off-campus stargazing events, hosted faculty research talks, and coordinated a 2024 solar eclipse viewing event for approximately 100 students; mentored board members to support long-term club sustainability.
- **Media Chair and Website Designer** *Sep 2020 – Jun 2023*
  Society of Physics Students, UC San Diego, La Jolla, CA
  ◦ Led a four-member media team, delegating tasks and designing promotional materials to support club events and improve student engagement.
  ◦ Built the club website from scratch, managed its online presence and social media, and initiated in-person event promotion in lower- and upper-division physics and mathematics courses.

Teaching Experience

- **Contract Tutor** *Sep 2024 – Dec 2024*
  Unlimited Learning, La Jolla, CA
  ◦ Tutored college students in mathematics and physics, providing individualized instruction in complex concepts.
  ◦ Developed tailored lesson plans and practice problems in vector calculus, linear algebra, and classical mechanics to reinforce understanding and prepare students for exams.
- **Supplemental Instruction Leader** *May 2022 – Dec 2022*
  Teaching + Learning Commons, UC San Diego, La Jolla, CA
  ◦ Led collaborative problem-solving sessions for up to 20 students in Vector Calculus I and II (MATH 20A, 20B, and 20C).
  ◦ Prepared weekly problem sets and reviewed course material to facilitate discussions and strengthen student understanding.

Memberships

- **American Astronomical Society (AAS)** *2022 - Present*
- **American Physical Society (APS)** *2020 - Present*

REFERENCES

1. **Michael W. Coughlin, PhD**
   Associate Professor, School of Physics and Astronomy
   University of Minnesota, Twin Cities
   Email: cough052@umn.edu
   Relationship: Current PhD Advisor

2. **Erik Katsavounidis, PhD**
   Senior Research Scientist, MIT LIGO Laboratory
   MIT Kavli Institute for Astrophysics and Space Research
   Email: kats@mit.edu
   Relationship: Gravitational-wave astrophysics and machine-learning research

3. **Deep Chatterjee, PhD**
   Research Scientist, MIT LIGO Laboratory
   MIT Kavli Institute for Astrophysics and Space Research
   Email: deep1018@mit.edu
   Relationship: Gravitational-wave astrophysics and machine-learning research

4. **Javier Duarte, PhD**
   Associate Professor, Department of Physics
   University of California, San Diego
   Email: jduarte@physics.ucsd.edu
   Relationship: Machine-learning research and A3D3 fellowship

5. **Peter Senchyna, PhD**
   Staff Astronomer
   Carnegie Observatories
   Email: psenchyna@carnegiescience.edu
   Relationship: CASSI research advisor